
USA
TODAY



UC Berkeley
Teaching Professor
Dan Garcia

CS61C

Great Ideas in Computer Architecture

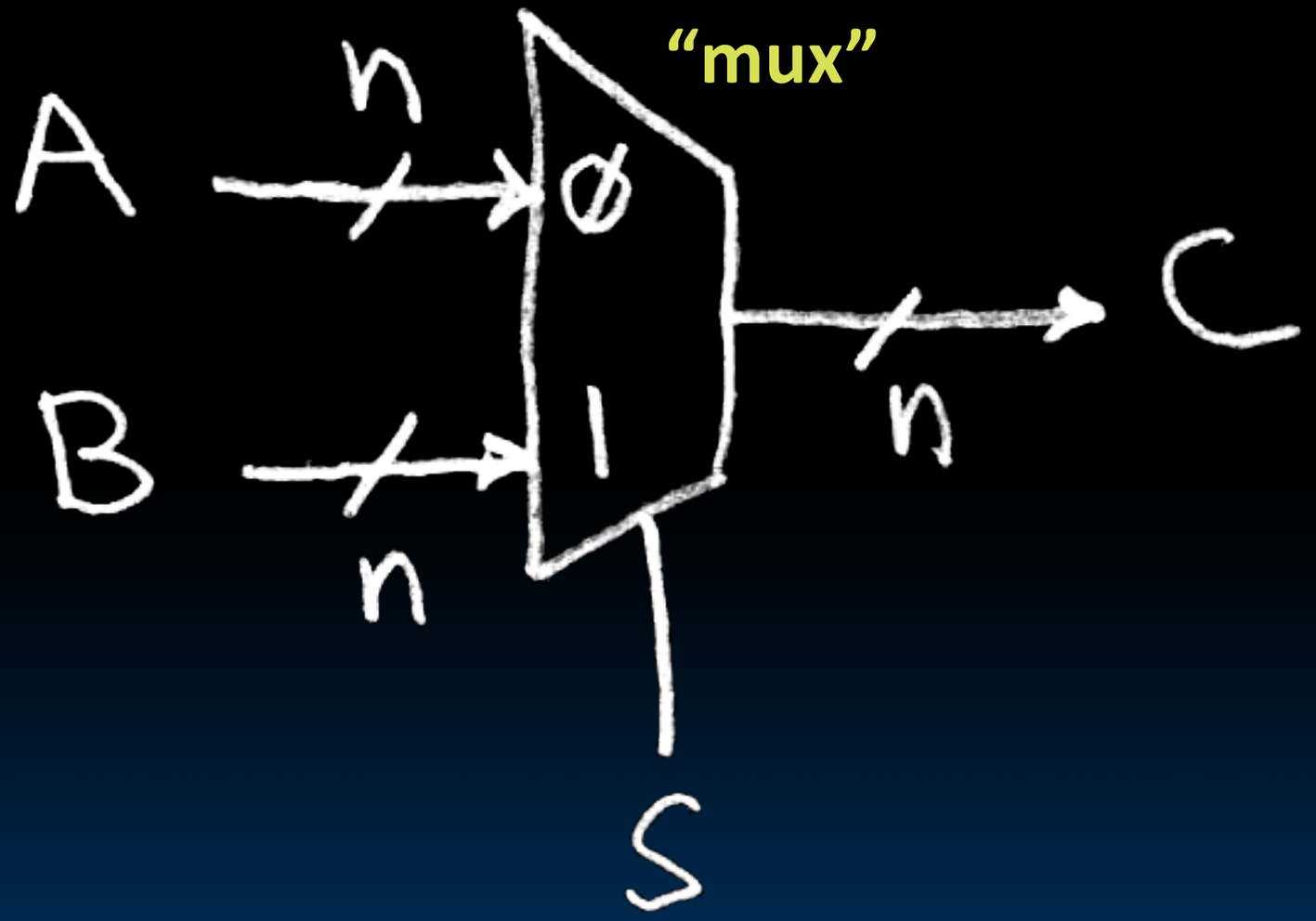
(a.k.a. Machine Structures)

Combinational Logic Blocks

BY NC SA

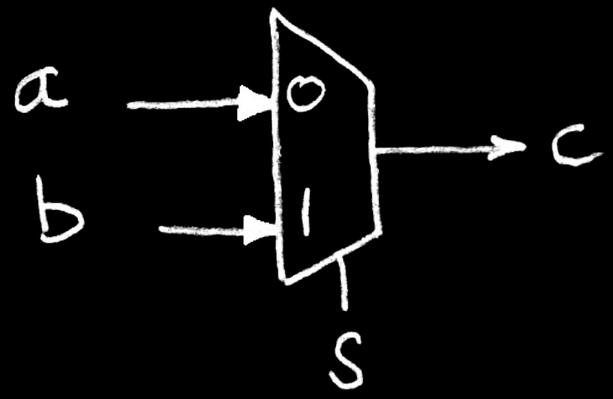
Data Multiplexors

Data Multiplexor (here 2-to-1, n-bit-wide)



BY NC SA

N instances of 1-bit-wide mux



How many rows in TT?

s	ab	c
0	00	0
0	01	0
0	10	1
0	11	1
1	00	0
1	01	1
1	10	0
1	11	1

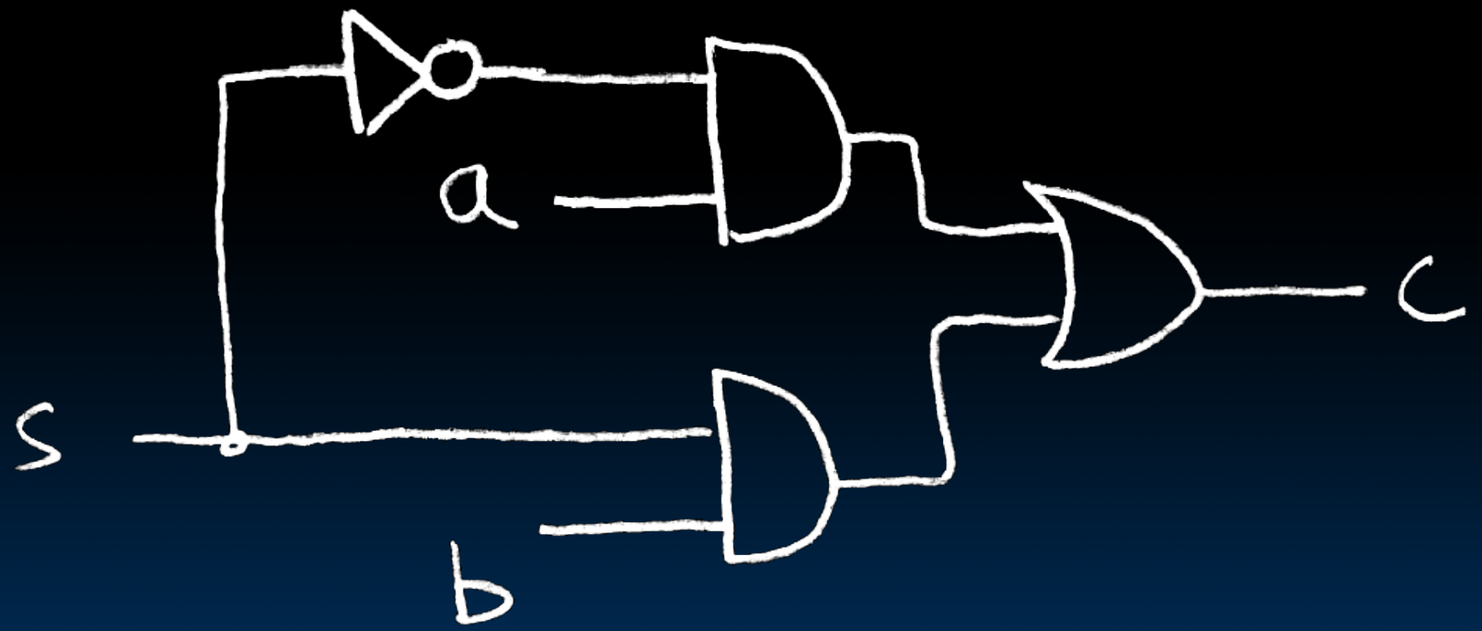
$$
\begin{aligned}
c &= \overline{s}a\overline{b} + \overline{s}ab + s\overline{a}b + sab \\
&= \overline{s}(a\overline{b} + ab) + s(\overline{a}b + ab) \\
&= \overline{s}(a(\overline{b} + b)) + s((\overline{a} + a)b) \\
&= \overline{s}(a(1) + s((1)b) \\
&= \overline{s}a + sb
\end{aligned}
$$

s	c
0	a
1	b

CC BY NC SA

How do we build a 1-bit-wide mux?

$$\overline{s}a + sb$$

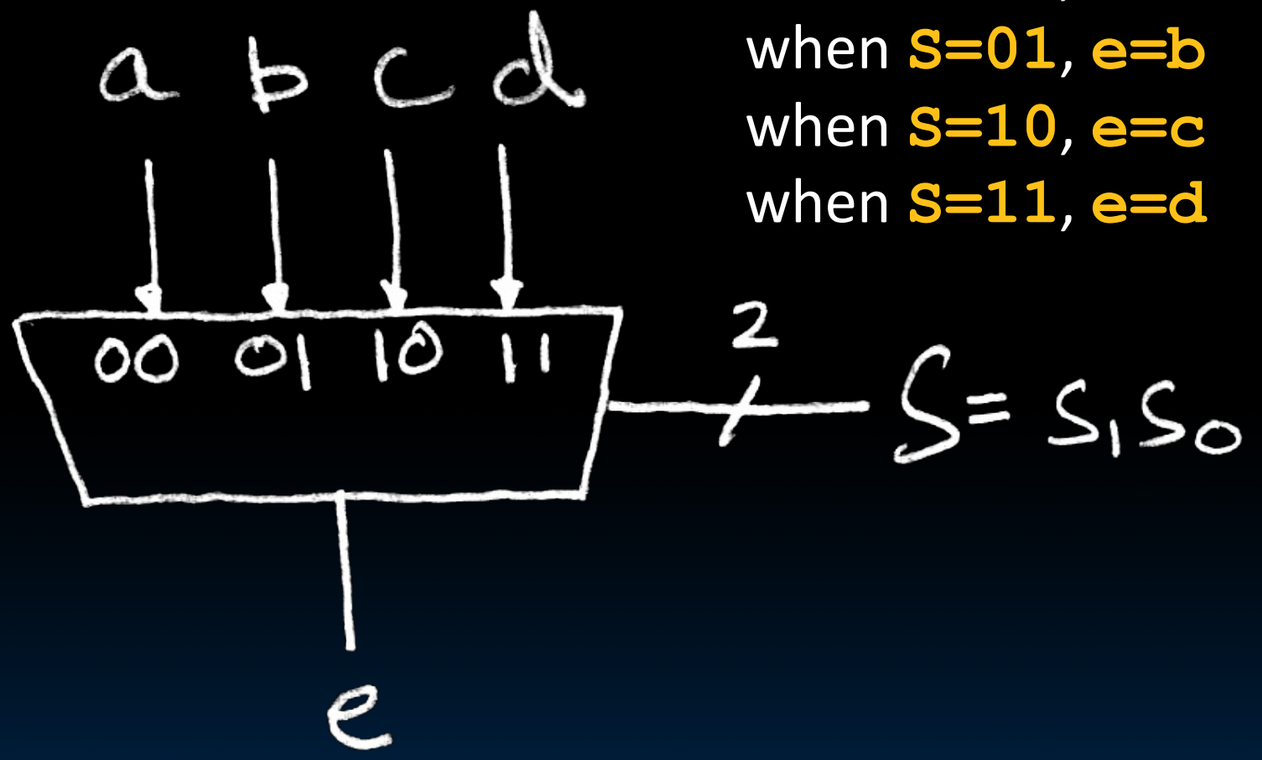


BY NC SA

4-to-1 Multiplexor?

- How many rows in the Truth Table?

when **S=00**, **e=a**
when **S=01**, **e=b**
when **S=10**, **e=c**
when **S=11**, **e=d**



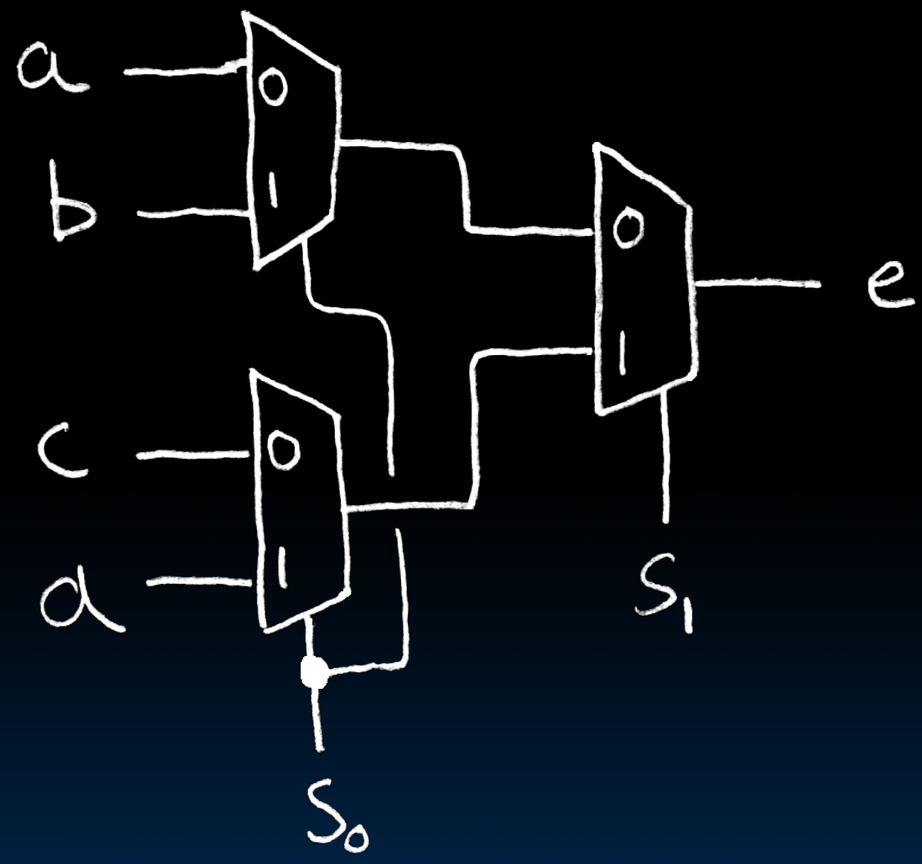
$s_1 s_0$	c
0 0	a
0 1	b
1 0	c
1 1	d

$$e = \overline{s_1} \cdot \overline{s_0} a + \overline{s_1} s_0 b + s_1 \overline{s_0} c + s_1 s_0 d$$

CC BY NC SA

Mux: is there any other way to do it?

Hint: NCAA tourney!



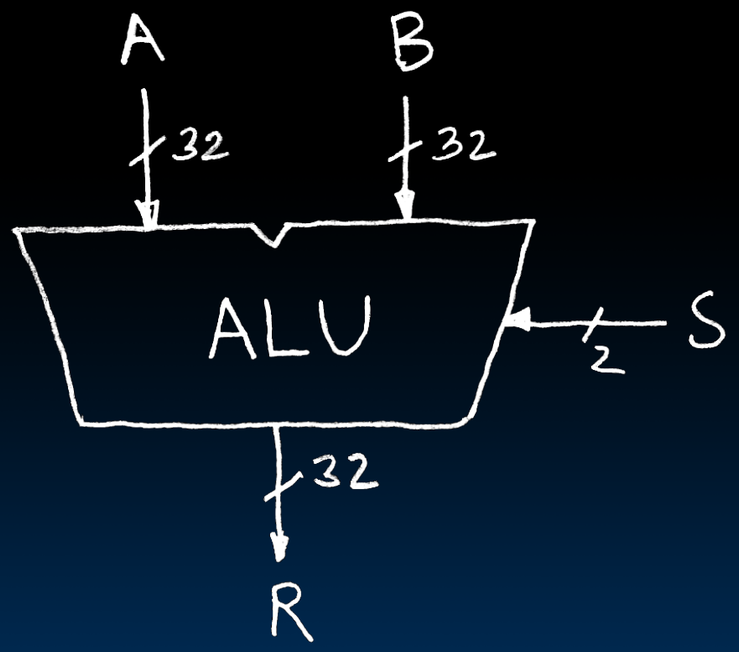
Ans: Hierarchically!

BY NC SA

Arithmetic Logic Unit (ALU)

Arithmetic and Logic Unit

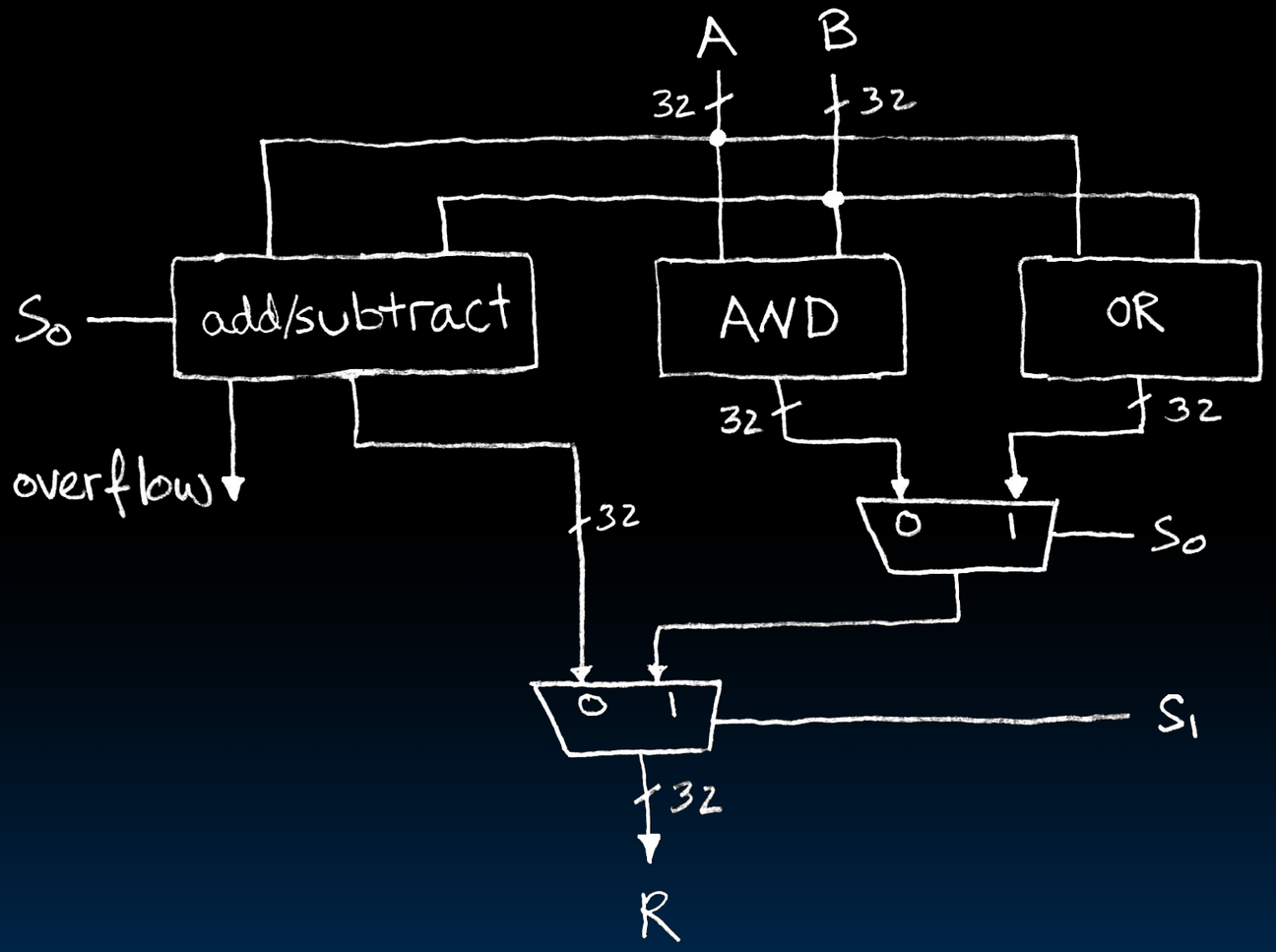
- Most processors contain a special logic block called "Arithmetic and Logic Unit" (ALU)
- We'll show you an easy one that does ADD, SUB, bitwise AND (**&**), bitwise OR (**|**)



when **S=00**, **R=A+B**
when **S=01**, **R=A-B**
when **S=10**, **R=A&B**
when **S=11**, **R=A|B**

BY NC SA

Our simple ALU





BY NC SA

Adder / Subtractor



Adder / Subtracter Design – how?

- Truth-table, then determine canonical form, then minimize and implement as we've seen before
- Look at breaking the problem down into smaller pieces that we can cascade or hierarchically layer

BY NC SA

Adder / Subtractor – One-bit adder LSB…

$$
\begin{array}{cccc|c}
 & a_3 & a_2 & a_1 & a_0 \\
+ & b_3 & b_2 & b_1 & b_0 \\
\hline
 & s_3 & s_2 & s_1 & s_0
\end{array}
$$

a_0	b_0	s_0	c_1
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

$$s_0 = a_0 \text{ XOR } b_0$$
$$c_1 = a_0 \text{ AND } b_0$$

CC BY NC SA

Adder / Subtractor – One-bit adder (1/2)…

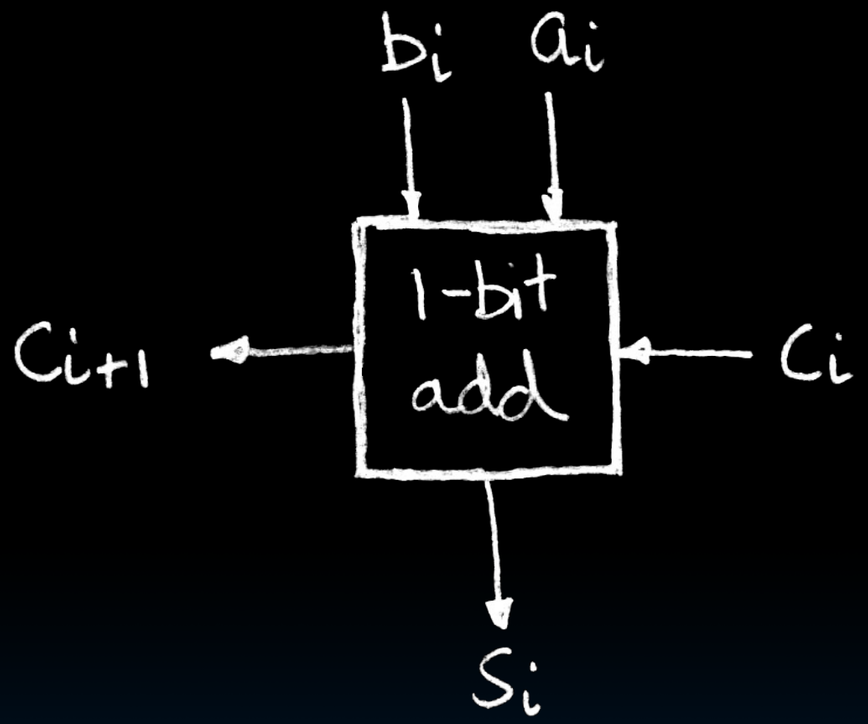
$$\begin{array}{cccccc} & a_3 & a_2 & \boxed{a_1} & a_0 \\ + & b_3 & b_2 & \boxed{b_1} & b_0 \\ \hline & s_3 & s_2 & \boxed{s_1} & s_0 \end{array}$$

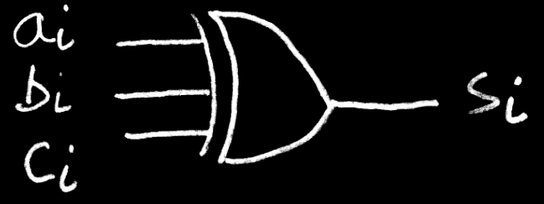
a_i	b_i	c_i	s_i	c_{i+1}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

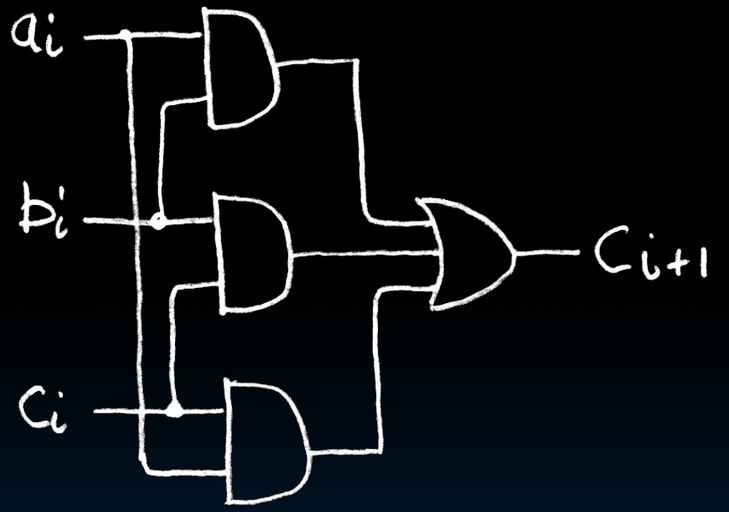
$$\begin{aligned} s_i &= \mathrm{XOR}(a_i, b_i, c_i) \\ c_{i+1} &= \mathrm{MAJ}(a_i, b_i, c_i) = a_i b_i + a_i c_i + b_i c_i \end{aligned}$$

CC BY NC SA

Adder / Subtractor – One-bit adder (2/2)…



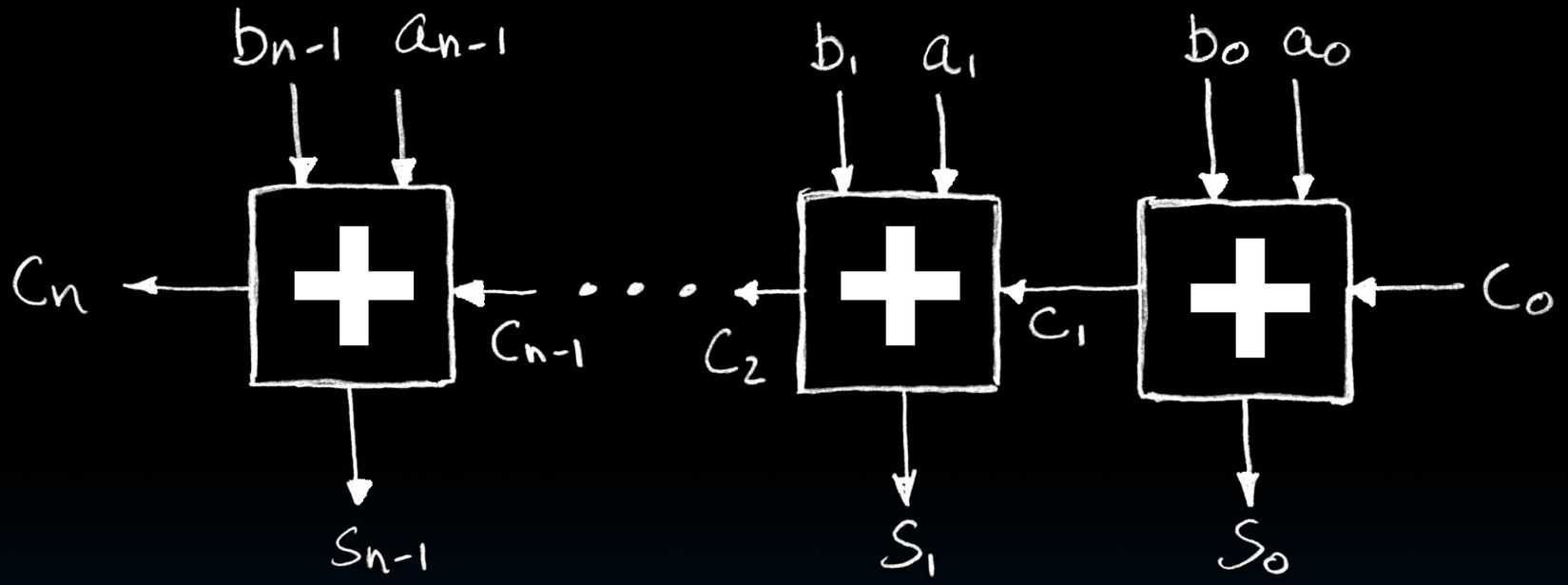




$$s_i = \mathbf{XOR}(a_i, b_i, c_i)$$
$$c_{i+1} = \mathbf{MAJ}(a_i, b_i, c_i) = a_i b_i + a_i c_i + b_i c_i$$

CC BY NC SA

N 1-bit adders → 1 N-bit adder

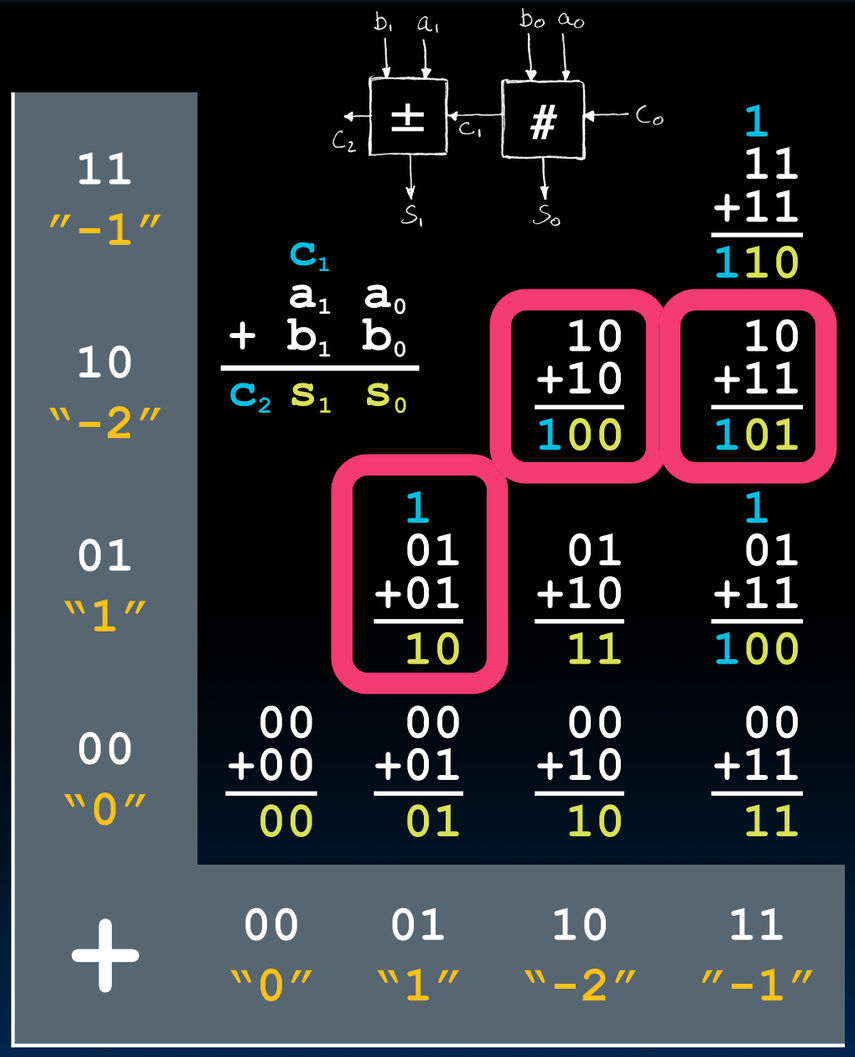


What about overflow?

Overflow = c_n?

CC BY NC SA

Sum of two 2-bit numbers…



- Let's add
  - First unsigned
  - Then signed (Two's Complement)
    - When do the lowest 2 bits of sum not represent correct sum?
    - Is there a pattern of when this happens? Hint: Check out the carry-bit and the sum-4s-column-bit
- Highest adder
  - C_{in} = Carry-in = c_1, C_{out} = Carry-out = c_2
  - No C_{out} or C_{in} → NO overflow!
  - C_{in} and C_{out} → NO overflow!
  - C_{in}, but no C_{out} → A,B both > 0, overflow!
  - C_{out}, but no C_{in} → A or B are -2, overflow!

What operation is this?

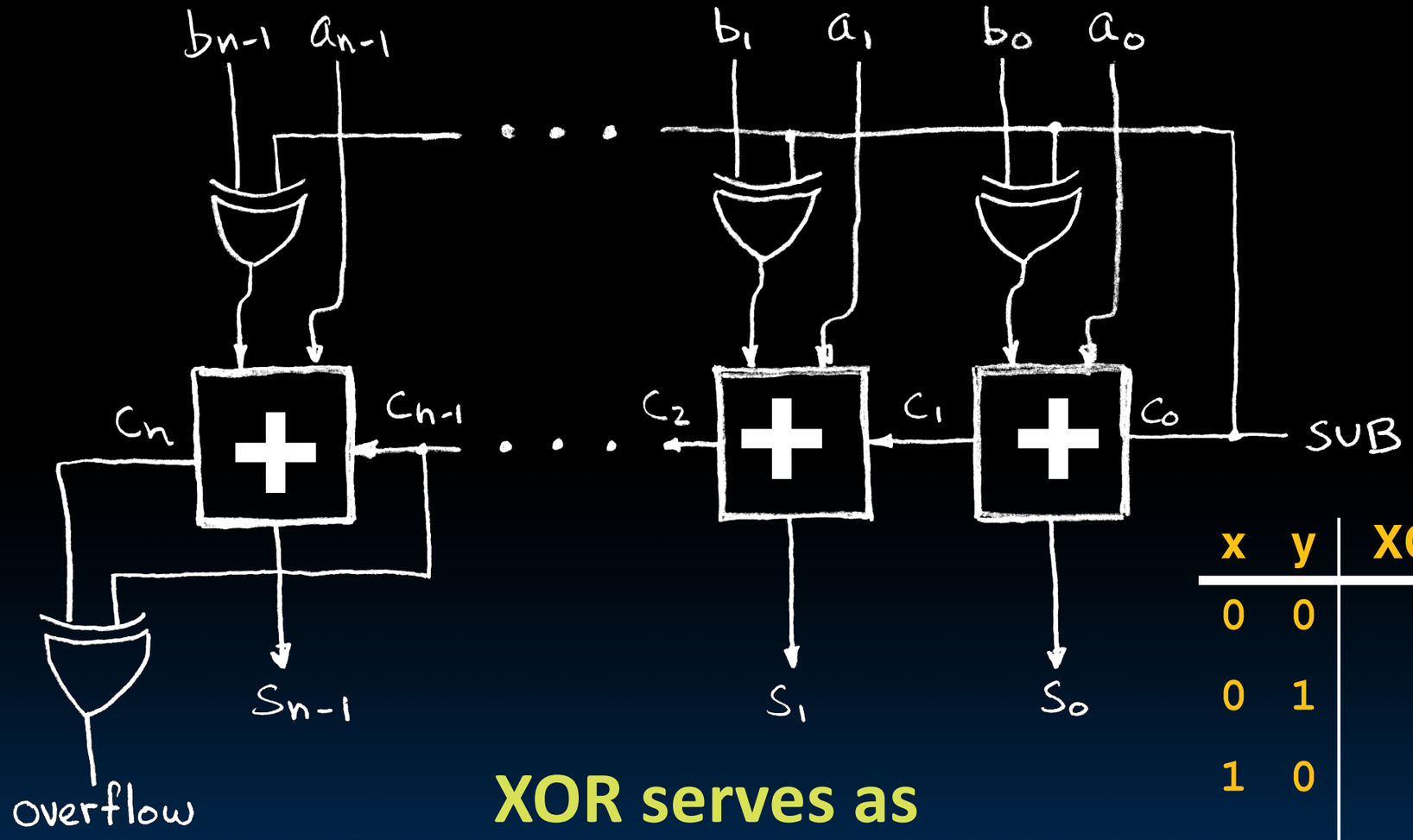
$$\text{overflow} = c_n \text{ XOR } c_{n-1}$$



CC BY NC SA

Subtractor Design

Extremely Clever Subtractor: A-B = A + (-B)



x	y	XOR(x,y)
0	0	0
0	1	1
1	0	1
1	1	0

XOR serves as conditional inverter!

CC BY NC SA

When poll is active, respond at **pollev.com/ddg**

Text **DDG** to **22333** once to join

L17

Truth table for mux with 4-bits of signals has 2^4 rows

We could cascade N 1-bit shifters to make 1 N-bit shifter for sll, srl

F	F
F	T
T	F
T	T

Powered by Poll Everywhere

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Answer

A. Truth table for mux with 4-bits of signals controls 16 inputs, for a total of 20 inputs, so truth table is 2^{20} rows…FALSE

B. We could cascade N 1-bit shifters to make 1 N-bit shifter for sll, srl … TRUE

A. Truth table for mux with 4-bits of signals has 2^4 rows

B. We could cascade N 1-bit shifters to make 1 N-bit shifter for sll, srl

AB
FF
FT
TF
TT

CC BY NC SA

“And In conclusion…”

- Use muxes to select among input
  - S input bits selects 2^S inputs
  - Each input can be n-bits wide, indep of S
- Can implement muxes hierarchically
- ALU can be implemented using a mux
  - Coupled with basic block elements
- N-bit adder-subtractor done using N 1-bit adders with XOR gates on input
  - XOR serves as conditional inverter



BY NC SA